

IDSS Linear Algebra In-Class Exercises

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16 October 2025

Note: *These practice questions have been designed to help you prepare for the IDSS exam, focusing on the Linear Algebra part of the module. Keep in mind that the actual exam questions will be different in content, style, and length, and they may even cover a different part of the taught material altogether. Please use these exercises to understand the material and practice the concepts and methods and do not memorise them.*

Questions

1. Assume the linear transformation described by a matrix M maps two points $P \rightarrow P'$ and $Q \rightarrow Q'$, such that $P' = MP$ and $Q' = MQ$.

Explain how you would calculate the matrix M from the coordinates of these two points and their transformations using linear algebra. Explain your method and reasoning, also providing all relevant equations and matrices that you would use for this calculation.

Solution: We are given two point correspondences:

$$P' = MP, \quad Q' = MQ$$

where P, Q are original points and P', Q' are transformed points. Assume these points are in 2D, so:

$$P = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}, \quad Q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \quad P' = \begin{bmatrix} p'_1 \\ p'_2 \end{bmatrix}, \quad Q' = \begin{bmatrix} q'_1 \\ q'_2 \end{bmatrix}$$

and the unknown matrix M is:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Formulation of the equations:

From $P' = MP$ and $Q' = MQ$, we have:

$$\begin{cases} ap_1 + bp_2 = p'_1 \\ cp_1 + dp_2 = p'_2 \\ aq_1 + bq_2 = q'_1 \\ cq_1 + dq_2 = q'_2 \end{cases}$$

Express as a matrix equation

Group the unknowns a, b, c, d into a vector:

$$\begin{bmatrix} p_1 & p_2 & 0 & 0 \\ 0 & 0 & p_1 & p_2 \\ q_1 & q_2 & 0 & 0 \\ 0 & 0 & q_1 & q_2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} p'_1 \\ p'_2 \\ q'_1 \\ q'_2 \end{bmatrix}$$

Let:

$$A = \begin{bmatrix} p_1 & p_2 & 0 & 0 \\ 0 & 0 & p_1 & p_2 \\ q_1 & q_2 & 0 & 0 \\ 0 & 0 & q_1 & q_2 \end{bmatrix}, \quad x = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}, \quad b = \begin{bmatrix} p'_1 \\ p'_2 \\ q'_1 \\ q'_2 \end{bmatrix}$$

So the system is:

$$Ax = b$$

Solve for x

If A is invertible (points are not collinear – A is non-singular), then:

$$x = A^{-1}b$$

Finally, reshape x into:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Python Example

```
# Define p* and q* here...
```

```
A = np.array([[p1, p2, 0, 0],
               [0, 0, p1, p2],
               [q1, q2, 0, 0],
               [0, 0, q1, q2]])
```

```
b = np.array([p1_prime, p2_prime, q1_prime, q2_prime])
```

```
x = np.linalg.solve(A, b)
```

```
M = x.reshape(2, 2)
```

2. Calculate M for $P = (5, 2)$, $P' = (-3, 4)$, $Q = (-1, -1)$ and $Q' = (2, 4)$.

Hint: Use row reduction (transformation to a row echelon form, or even to a reduced row echelon form – RREF) to achieve this.

Solution:

We start with the augmented matrix for the system:

$$\left[\begin{array}{cccc|c} 5 & 2 & 0 & 0 & -3 \\ 0 & 0 & 5 & 2 & 4 \\ -1 & -1 & 0 & 0 & 2 \\ 0 & 0 & -1 & -1 & 4 \end{array} \right]$$

Divide row 1 by 5:

$$R_1 \leftarrow \frac{1}{5}R_1$$

$$\left[\begin{array}{cccc|c} 1 & \frac{2}{5} & 0 & 0 & -\frac{3}{5} \\ 0 & 0 & 5 & 2 & 4 \\ -1 & -1 & 0 & 0 & 2 \\ 0 & 0 & -1 & -1 & 4 \end{array} \right]$$

Eliminate below pivot in column 1:

$$R_3 \leftarrow R_3 + R_1$$

$$\left[\begin{array}{cccc|c} 1 & \frac{2}{5} & 0 & 0 & -\frac{3}{5} \\ 0 & 0 & 5 & 2 & 4 \\ 0 & -\frac{3}{5} & 0 & 0 & \frac{7}{5} \\ 0 & 0 & -1 & -1 & 4 \end{array} \right]$$

Make pivot in row 3 equal to 1 – divide row 3 by $-\frac{3}{5}$:

$$R_3 \leftarrow -\frac{5}{3}R_3$$

$$\left[\begin{array}{cccc|c} 1 & \frac{2}{5} & 0 & 0 & -\frac{3}{5} \\ 0 & 0 & 5 & 2 & 4 \\ 0 & 1 & 0 & 0 & -\frac{7}{3} \\ 0 & 0 & -1 & -1 & 4 \end{array} \right]$$

Eliminate above pivot in column 2:

$$R_1 \leftarrow R_1 - \frac{2}{5}R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 5 & 2 & 4 \\ 0 & 1 & 0 & 0 & -7/3 \\ 0 & 0 & -1 & -1 & 4 \end{array} \right]$$

Make pivot in row 2 equal to 1 – divide row 2 by 5:

$$R_2 \leftarrow \frac{1}{5}R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 1 & 2/5 & 4/5 \\ 0 & 1 & 0 & 0 & -7/3 \\ 0 & 0 & -1 & -1 & 4 \end{array} \right]$$

Eliminate below pivot in column 3:

$$R_4 \leftarrow R_4 + R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 1 & 2/5 & 4/5 \\ 0 & 1 & 0 & 0 & -7/3 \\ 0 & 0 & 0 & -3/5 & 24/5 \end{array} \right]$$

Make pivot in row 4 equal to 1 – divide row 4 by $-3/5$:

$$R_4 \leftarrow -\frac{5}{3}R_4$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 1 & 2/5 & 4/5 \\ 0 & 1 & 0 & 0 & -7/3 \\ 0 & 0 & 0 & 1 & -8 \end{array} \right]$$

(Note: By exchanging rows 2 and 3, above, we can reach a row echelon form and can easily solve by back substitution, i.e., starting by R_4 and moving upwards. Below, we demonstrate solving the system by reducing the augmented matrix to its reduced row echelon form (RREF), where all entries outside the pivot positions and the augmented column are zero.)

Eliminate above pivot in column 4:

$$R_2 \leftarrow R_2 - \frac{2}{5}R_4$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 1 & 0 & 4 \\ 0 & 1 & 0 & 0 & -7/3 \\ 0 & 0 & 0 & 1 & -8 \end{array} \right]$$

Final reduced row echelon form (RREF):

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1/3 \\ 0 & 1 & 0 & 0 & -7/3 \\ 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & -8 \end{array} \right]$$

Thus:

$$a = \frac{1}{3}, \quad b = -\frac{7}{3}, \quad c = 4, \quad d = -8$$

$$M = \begin{bmatrix} 1/3 & -7/3 \\ 4 & -8 \end{bmatrix}$$

Verification:

$$MP = \begin{bmatrix} 1/3 & -7/3 \\ 4 & -8 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}, \quad MQ = \begin{bmatrix} 1/3 & -7/3 \\ 4 & -8 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

3. Solve the system above for $P = (2, 4)$, $P' = (1, 1)$, $Q = (4, 8)$, $Q' = (2, 2)$. What can be said about the (linear transformation) matrix M in this case?

Solution: We have:

$$P = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad Q = \begin{bmatrix} 4 \\ 8 \end{bmatrix}, \quad P' = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad Q' = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

The augmented matrix is:

$$\left[\begin{array}{cccc|c} 2 & 4 & 0 & 0 & 1 \\ 0 & 0 & 2 & 4 & 1 \\ 4 & 8 & 0 & 0 & 2 \\ 0 & 0 & 4 & 8 & 2 \end{array} \right]$$

Make pivot in row 1 equal to 1 – divide row 1 by 2:

$$R_1 \leftarrow \frac{1}{2}R_1$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 2 & 4 & 1 \\ 4 & 8 & 0 & 0 & 2 \\ 0 & 0 & 4 & 8 & 2 \end{array} \right]$$

Eliminate below pivot in column 1:

$$R_3 \leftarrow R_3 - 4R_1$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 2 & 4 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 8 & 2 \end{array} \right]$$

(Notice that $R_3 = \mathbf{0}$. This shows that R_3 was a linear combination of R_1 . Therefore the coefficient matrix A is singular i.e., $\det(A) = 0$, and it cannot be inverted.

Strictly speaking we don't need to continue, however we do for completeness and to determine how many rows are linearly dependent on other rows.)

Make pivot in row 2 equal to 1 – divide row 2 by 2:

$$R_2 \leftarrow \frac{1}{2}R_2$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 8 & 2 \end{array} \right]$$

Eliminate below pivot in column 3:

$$R_4 \leftarrow R_4 - 4R_2$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

The final RREF:

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Interpretation: Pivot columns are C_1 and C_3 . C_2 and C_4 are free variables. Therefore, the system does not have a unique solution because the original points are linearly dependent.

4. Assuming a square matrix $M = [m_{ij}] = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ show the relationship between m_{ij} of matrix M that indicate the M is singular.

Hint: Start by assuming a linear dependence between the rows of M , which indicates that the implied system has no or not unique solutions.

Solution: For M to be singular, it must be that there is a scalar k such that $R_1 = kR_2$. Therefore:

$$\begin{aligned} \begin{bmatrix} a & b \end{bmatrix} &= k \begin{bmatrix} c & d \end{bmatrix} \\ &= \begin{bmatrix} kc & kd \end{bmatrix} \end{aligned}$$

It follows that $k = \frac{a}{c}$ and that $k = \frac{b}{d}$, therefore that

$$\begin{aligned} \frac{a}{c} &= \frac{b}{d} \\ \leadsto ad &= bc \\ \leadsto ad - bc &= 0 \end{aligned}$$

which is how we define the determinant of a square matrix:

$$\det(M) = ad - bc$$

The determinant then *determines whether M is singular*. If $\det(M) = 0$, then M is singular (and non-invertible), meaning its rows (and columns) are linearly dependent. It follows that the associated

system of linear equations does not have a unique solution; it may have no solution or infinitely many solutions. Geometrically, in two dimensions, the lines represented by these equations are either parallel (no solution) or coincident (infinitely many solutions).

5. The rank of a matrix is the maximum number of linearly independent rows or columns. It tells us the dimension of the row space or column space – that is, the number of directions in which the matrix contains meaningful information.

Compute the rank of the matrix below, using row reduction:

$$A = \begin{bmatrix} 37 & -22 & 59 & 81 \\ 111/2 & -33 & 177/2 & 243/2 \\ -74 & 44 & -118 & -162 \\ 185/4 & -110/4 & 295/4 & 405/4 \end{bmatrix}$$

Solution:

Use R_1 as the pivot row. Apply the following row operations:

$$R_2 \leftarrow R_2 - \frac{3}{2}R_1,$$

$$R_3 \leftarrow R_3 + 2R_1,$$

$$R_4 \leftarrow R_4 - \frac{5}{4}R_1.$$

After these operations:

$$\begin{bmatrix} 37 & -22 & 59 & 81 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

No further reduction is needed. Only one non-zero row remains. It follows that

$$\text{Rank}(A) = 1$$

Therefore, all rows are scalar multiples of the first row, so the row space is 1-dimensional. The matrix acts like a projection or scaling along one direction, so it effectively squashes 4D space into a 1D line.

6. Given a zero-mean dataset represented by the matrix

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

compute the *whitening transformation matrix* W such that the transformed data $Z = WX$ has an identity covariance matrix.

Hint: Use the eigendecomposition of the covariance matrix.

Solution:

First, we compute the covariance matrix. Since X is zero-centred or zero-mean:

$$\Sigma = XX^T = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

Then, we perform eigen-decomposition on Σ .

Since Σ is diagonal, its eigenvectors are the standard basis vectors:

$$u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

(This is because a diagonal matrix does not rotate, it only scales. Because a diagonal matrix acts purely as a scaling operation along the standard coordinate axes, those axes naturally become the special, unrotated directions – eigenvectors.)

Arranging them as columns gives:

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

The eigenvalues are: $\lambda_1 = 4, \lambda_2 = 1$.

Now we can compute the whitening matrix:

$$W = \Lambda^{-\frac{1}{2}} U^T = \begin{bmatrix} \frac{1}{\sqrt{4}} & 0 \\ 0 & \frac{1}{\sqrt{1}} \end{bmatrix} = \begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix}$$

Verification:

$$\text{Cov}(Z) = W\Sigma W^T = I$$

7. Consider the covariance matrix

$$\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

Compute the whitening matrix W using eigen-decomposition.

(Whitening transformation – general case.)

Solution:

Step 1: Eigen-decomposition of Σ . Compute eigenvalues by solving:

$$\det(\Sigma - \lambda I) = \begin{vmatrix} 4 - \lambda & 2 \\ 2 & 3 - \lambda \end{vmatrix} = (4 - \lambda)(3 - \lambda) - 4 = \lambda^2 - 7\lambda + 8$$

$$\lambda_{1,2} = \frac{7 \pm \sqrt{49 - 32}}{2} = \frac{7 \pm \sqrt{17}}{2} \approx 5.56, 1.44$$

Step 2: Compute eigenvectors. For $\lambda_1 \approx 5.56$:

$$(\Sigma - \lambda_1 I)v = 0 \implies v_1 \propto \begin{bmatrix} 0.85 \\ 0.53 \end{bmatrix}$$

For $\lambda_2 \approx 1.44$:

$$v_2 \propto \begin{bmatrix} -0.53 \\ 0.85 \end{bmatrix}$$

Normalize to get:

$$U = \begin{bmatrix} 0.85 & -0.53 \\ 0.53 & 0.85 \end{bmatrix}$$

Step 3: Compute $\Lambda^{-\frac{1}{2}}$:

$$\Lambda^{-\frac{1}{2}} = \begin{bmatrix} 1/\sqrt{5.56} & 0 \\ 0 & 1/\sqrt{1.44} \end{bmatrix} \approx \begin{bmatrix} 0.42 & 0 \\ 0 & 0.83 \end{bmatrix}$$

Step 4: The whitening matrix is then:

$$W = \Lambda^{-\frac{1}{2}} U^T$$

This rotates the data to the eigenbasis and rescales it so that the covariance becomes the identity.

8. Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

and interpret their geometric meaning.

Solution:

Step 1: First solve the characteristic polynomial to obtain the eigenvalues:

$$\det(A - \lambda I) = \begin{vmatrix} 3 - \lambda & 1 \\ 0 & 2 - \lambda \end{vmatrix} = (3 - \lambda)(2 - \lambda)$$

$$(3 - \lambda)(2 - \lambda) = 0 \implies \lambda_1 = 3, \lambda_2 = 2$$

Step 2: Find the eigenvectors: For $\lambda_1 = 3$:

$$(A - 3I)v = 0 \implies \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} v = 0 \implies v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

For $\lambda_2 = 2$:

$$(A - 2I)v = 0 \implies \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} v = 0 \implies v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

The matrix scales vectors along v_1 by 3 and along v_2 by 2. Geometrically, A stretches space differently along these two directions.

9. Compute the singular value decomposition of

$$B = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$$

and verify that $B = U\Sigma V^T$.

Solution: Step 1: Compute $B^T B$.

$$B^T B = \begin{bmatrix} 0 & 4 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 0 & 9 \end{bmatrix}$$

Step 2: Compute singular values. Eigenvalues of $B^T B$ are 16 and 9. Singular values: $\sigma_1 = 4, \sigma_2 = 3$. So

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$$

Step 3: Compute V . Since $B^T B$ is diagonal,

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Step 4: Compute U .

$$U = BV\Sigma^{-1} = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 1/4 & 0 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Step 5: Verify.

$$B = U\Sigma V^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$$

10. Prove that the eigenvalues of a diagonal matrix are the elements on its main diagonal. Then, explain how these relate to the singular values of the matrix.

Solution: Step 1: Define a diagonal matrix. Let

$$D = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix}$$

Step 2: Compute the characteristic polynomial.

$$\det(D - \lambda I) = \det \begin{bmatrix} d_1 - \lambda & 0 & \dots & 0 \\ 0 & d_2 - \lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n - \lambda \end{bmatrix}$$

The determinant of a diagonal matrix is the product of its diagonal entries:

$$\det(D - \lambda I) = (d_1 - \lambda)(d_2 - \lambda) \dots (d_n - \lambda)$$

Step 3: Solve for eigenvalues. The roots of the characteristic polynomial are:

$$\lambda_i = d_i \quad \text{for } i = 1, 2, \dots, n$$

Thus, the eigenvalues of a diagonal matrix are exactly its diagonal entries.

Step 4: Relationship to singular values. The singular values of any matrix A are the square roots of the eigenvalues of $A^\top A$. For a diagonal matrix D , we have:

$$D^\top D = \begin{bmatrix} d_1^2 & 0 & \dots & 0 \\ 0 & d_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n^2 \end{bmatrix}$$

The eigenvalues of $D^\top D$ are $d_1^2, d_2^2, \dots, d_n^2$. Therefore, the singular values of D are:

$$\sigma_i = |d_i| \quad \text{for } i = 1, 2, \dots, n$$

Interpretation: For a diagonal matrix, the eigenvalues are the diagonal entries themselves, which may be positive, negative, or zero. Geometrically, these eigenvalues describe how the matrix scales vectors along the coordinate axes:

- If $d_i > 0$, the i -th axis is stretched by d_i .
- If $d_i < 0$, it is stretched by $|d_i|$ and reflected through the origin.
- If $d_i = 0$, the axis collapses to zero.

Singular values measure the magnitude of stretching without regard to sign. They are always non-negative and equal to $|d_i|$. Geometrically, applying D to the unit sphere produces an axis-aligned ellipsoid whose semi-axes have lengths $|d_1|, |d_2|, \dots, |d_n|$. Thus:

Eigenvalues: signed scaling factors; Singular values: absolute scaling factors.

Example: Consider $D = \text{diag}(3, -2, 0)$.

- Eigenvalues: $3, -2, 0$ (stretch by 3 along x , reflect and stretch by 2 along y , collapse z).
- Singular values: $|3|, |-2|, |0| = 3, 2, 0$ (ellipsoid with radii 3 and 2 along x and y axes).